



AI_CONCIERGE / WEEKLY_REPORT.md

AI Concierge

Weekly Progress Report

Technical Documentation • System Design • ML Architecture

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Report cycle: Weekly • Status: On Track

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01 Project Progress Overview

overview

This week established the technical foundation for the AI Concierge project. Effort was concentrated on two parallel tracks — backend architecture and documentation, and ML/AI documentation — with the goal of getting the project ready to move from planning into implementation.

- Established the technical foundation for the AI Concierge project.
- Focused this week on backend architecture/documentation and ML/AI documentation.
- Finalized a structured repository organization for documentation, implementation, infrastructure and testing.
- Prepared the project for the transition from documentation to implementation.

02 Backend Documentation Progress

backend_docs/

- Documented backend architecture and API design.
- Defined database design and ER-model considerations.
- Documented authentication and backend security considerations.
- Defined memory, RAG and agent integration points with the backend.

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Established implementation-oriented backend documentation under backend_docs/.

03 ML / AI Documentation Progress

ml/

- Documented ML architecture and data-processing considerations.
- Covered embeddings, retrieval, reranking and RAG.
- Documented LLM and prompt-engineering considerations.
- Included memory, recommendation, evaluation and AI-safety documentation areas.

i

Established the ML/AI documentation structure under ml/.

04 System Design Organization

docs/

Maintained separation between system-level design and implementation-level documentation, so that architectural intent and implementation detail can evolve independently.

docs/system_design/

Captures architectural decisions and overall system design.

backend_docs/

Captures backend-specific implementation documentation.

ml/

Captures ML and AI methodology.

docs/mlops/

Captures MLOps/LLMOps lifecycle and operational concerns.

05 Repository Structure Finalization

repo/

```
ai-concierge/  
|-- backend/  
|-- frontend/  
|-- infrastructure/  
|-- tests/  
|-- backend_docs/  
|-- ml/  
|-- docs/  
|   |-- system_design/  
|   |-- product/  
|   |-- development/  
|   |-- mlops/  
|-- Decision_Log.md
```

- Finalized a canonical repository structure.
- Separated backend, frontend, infrastructure and tests from documentation.
- Defined dedicated locations for product, system-design, development and MLOps documentation.
- Established **Decision_Log.md** as the location for future architectural changes.

06 Key Technical Areas Covered

coverage

Backend

APIs, authentication, database, services and integrations.

AI

LLMs, embeddings, RAG, retrieval and reranking.

Personalization

Memory architecture and recommendation-related components.

Engineering

Security, testing, monitoring, deployment and evaluation.

Operations

MLOps, LLMOps, versioning, CI/CD and AI monitoring.

07 Outcome of This Week

outcome

- A coherent technical documentation baseline is now established.
- Major backend and ML components have defined documentation boundaries.
- The project structure supports incremental implementation.
- The architecture can now be reviewed before committing to implementation details.



The project is ready to transition from design/documentation to development.

08 Next Steps

next/

- 1 Begin backend implementation using the finalized structure.
- 2 Set up the backend application skeleton and environment.
- 3 Implement database configuration and core models.

- 4 Implement authentication and initial API endpoints.
- 5 Build the first end-to-end application flow.
- 6 Then progressively integrate RAG, memory and agent components.

09 Mentor Review / Discussion Points

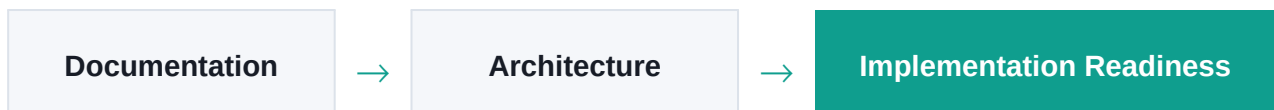
review/

Open items for discussion in the mentor meeting, in priority order:

- Validate the proposed backend and ML architecture.
- Review whether the separation between **backend_docs/**, **ml/** and **docs/** is appropriate.
- Identify architectural risks before implementation.
- Prioritize the first backend features to implement.
- Confirm the proposed implementation sequence and evaluation strategy.

10 Weekly Milestone

milestone



The major achievement this week is establishing a structured technical foundation.

The immediate objective is now to convert the documented design into a working prototype.

— End of Report —